Resourcing

Overview

Analyzing questions related to funding and constraints.

Research questions:
- How are OSPOs funded?
- What fraction of OSPO contributors are volunteers?
- What are the OSPOs' biggest blockers?

Relevant survey questions:
- How are your OSPO activities funded or allocated?
- Roughly how many people regularly spend time on OSPO activities without being formally resourced?
- Which of the following blockers has your OSPO encountered?

Import packages, utilities, and config variables

```
library(readxl)
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag
```

```
The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
library(tibble)
```

```
Warning: package 'tibble' was built under R version 4.4.3
```

```
library(docstring)
```

```
Attaching package: 'docstring'

The following object is masked from 'package:utils':

    ?
```

```
library(readr)
```

```
Warning: package 'readr' was built under R version 4.4.3
```

```
library(tidyr)
```

```
Warning: package 'tidyr' was built under R version 4.4.3
```

```
source("utils.R")

# Read config.yml
cfg <- config::get()
```

Load data

```
org_responses <- read_excel(file.path(cfg$curated_path, "orgs.xlsx"), sheet="responses")
codebook <- read_excel(file.path(cfg$data_path, "codebook.xlsx"))
mc_simple <- read_tsv(file.path(cfg$data_path, "mc_simple.tsv"), show_col_types = FALSE, ski

tools_infra <- read_tsv(file.path(cfg$data_path, "tools_infra.tsv"), show_col_types = FALSE,
ai_topics <- read_tsv(file.path(cfg$data_path, "ai_topics.tsv"), show_col_types = FALSE, ski
ai_usage <- read_tsv(file.path(cfg$data_path, "ai_usage.tsv"), show_col_types = FALSE, skip_
tools_reliance <- read_tsv(file.path(cfg$data_path, "tools_reliance.tsv"), show_col_types =
blockers <- read_tsv(file.path(cfg$data_path, "blockers.tsv"), show_col_types = FALSE, skip_
matrix_other <- read_tsv(file.path(cfg$data_path, "matrix_other.tsv"), show_col_types = FALS
sata_other <- read_tsv(file.path(cfg$data_path, "sata_other.tsv"), show_col_types = FALSE, s
```

What fraction of OSPO contributors are volunteers?

First, inspect answers from each org and check for disagreement from within the same org.

```
mc_simple$organization <- org_responses$organization
volunteers <- mc_simple |>
    select(volunteers, organization)

dupes <- volunteers |>
  group_by(organization) |>
  filter(n() > 1) |>
  arrange(organization, .by_group = TRUE) |>
  select(volunteers, organization)

# Keep commented for privacy!!
# dupes
```

Okay, great. I can see that only at one university did all respondents give a consistent answer. Similar to what I did in the "tools" notebook, I am going to just take the largest of the answers, because I expect that people are more likely to underestimate than overestimate on this one.

```
# Order of preference: keep whichever is further down the list
ordered_sizes <- c(
    "I don't know",
    "None",
    "1 person",
    "2 to 5 people",
    "6-10 people"
```

```
)

volunteers_consolidated <- volunteers |>
  mutate(volunteers = factor(volunteers, levels = ordered_sizes)) |>
  group_by(organization) |>
  slice_max(volunteers, n = 1, with_ties = FALSE) |>
  ungroup() |>
  mutate(volunteers = as.character(volunteers))


volunteers_consolidated |> count(volunteers)
```

```
# A tibble: 6 x 2
  volunteers        n
  <chr>         <int>
1 1 person          4
2 2 to 5 people    10
3 6-10 people       4
4 I don't know      3
5 None              5
6 <NA>              1
```